

Bathroom Planning Is More Than Choosing Fixtures

Bathrooms represent less than 10% of residential square footage, yet generate disproportionate renovation friction. Hidden plumbing offsets, illegal door swings, and mismatched catalog specs leave homeowners vulnerable to costly change orders and force trade designers into repetitive manual redrafts.

“Describe your space. Define your style. Explore possibilities. Understand the impact of every change. Finalize with confidence.”

FOR HOMEOWNERS

Eliminates surprise change orders; guarantees visual coherence and ergonomic clearance.

FOR TRADE DESIGNERS

Generates code-compliant 2D schematics, verified fixture pairings, and automated bill-of-materials.

THE GOLDEN PATH WORKFLOW (SELECT PHASE TO INSPECT)

STAGE 1 ACTIVE

01. BRIEF

Intent Intake

Dimensions, budget & style extraction.

02. DESIGN

Catalog Match

2D spatial layout & SKU alignment.

03. REFINE

DesignPulse™

Reactive downstream impact analysis.

04. FINALIZE

Build Spec

Code-audited package for construction.

Stage 01 Focus: Conversational intake extracts dimensions, user aesthetic preferences, and budget bounds into structured state.



Precision Wet-Room Engineering
Harmonizing high-craft aesthetic ambition with deterministic spatial clearances and plumbing rough-in constraints.

What If Every Design Change Came With Its Impact?

In wet environments, modifying a single fixture causes cascading spatial, hydraulic, and budgetary consequences.

Trigger: Swap 36" Single → 60" Double Vanity

USER MODIFICATION

User requests upgrade to **KOHLER Verdera 60" Dual Basin** to accommodate twin morning routines.

Constraint Violations Identified

⚠️ CLEARANCE CONFLICT

Door Swing Clearance: Drops to 18.5" (violates NKBA Guideline #4: 21" minimum required walkway).
Plumbing Rough-In Delta: Dual-waste manifold split requires opening wet wall (+\$380 trade labor).

DesignPulse™ Proactive Synthesis

Rather than failing silently or rejecting the change, DesignPulse™ computes two code-compliant alternatives above.

DesignPulse™ Workflow: Change Request → Impact Analysis → Affected/Unaffected Ledger → Trade-off Options → User Control.

Downstream Impact Ledger

STATE: CONFLICT

Subsystem Element	Status	Calculated Variance / Note
Entry Door Walkway	18.5" CLASH	Violates NKBA 21" min requirement
Plumbing Rough-In	MODIFIED	Dual waste split required (+\$380)
Walk-In Shower Zone	UNAFFECTED	Zero envelope or drain shift
Water Closet Alcove	UNAFFECTED	30" centerline rough-in intact
Wall Tile Consumption	UNAFFECTED	Net surface delta ±0 sq ft
Verified Budget Total	CALCULATED	\$9,840 (+\$1,420 net delta)



AI-Assisted. Deterministically Evaluated. Transparently Explained.

“AI imagines. Rules verify. AR visualizes.” Hallucinations are prevented through strict authority separation.



1. Conversational Ingestion

Authority: LLM Agent

Extracts room bounds, style language, and budget parameters from natural language into validated JSON project state.



2. Deterministic Layout Solver

Authority: Geometric Solver

Calculates fixture coordinate envelopes, door swing arcs, and minimum NKBA walkway clearances via rigid math.



3. DesignPulse™ Impact Engine

Authority: DAG Rules Graph

Evaluates upstream fixture shifts against hydraulic connections, spatial constraints, and live catalog price deltas.



4. Specification & Verification

Authority: Catalog & Human Pro

Binds verified KOHLER SKUs to 2D SVG schematics and flags unverified in-wall rough-ins for licensed contractor sign-off.

Source of Truth Matrix

Ensuring spatial, hydraulic, and contractual accuracy.

CONVERSATIONAL EXPLANATION

LLM Assistant

Subjective intent translation

GEOMETRY, CODE & BOM

Deterministic Rules

Mathematical proof

CONCEALED IN-WALL ROUGH-IN

Licensed Contractor

Mandatory on-site check



Confidence in Every Decision. Value in Every Footprint.

Measurable operational savings for trade designers; total financial certainty for homeowners.

\$2,500+
Rework Avoidance

Prevents post-purchase tear-outs caused by clearance clashes and rough-in errors.

4 Days » 15 Min
Schematic Turnaround

Compresses manual drafting cycles into an instant, verified client co-creation session.

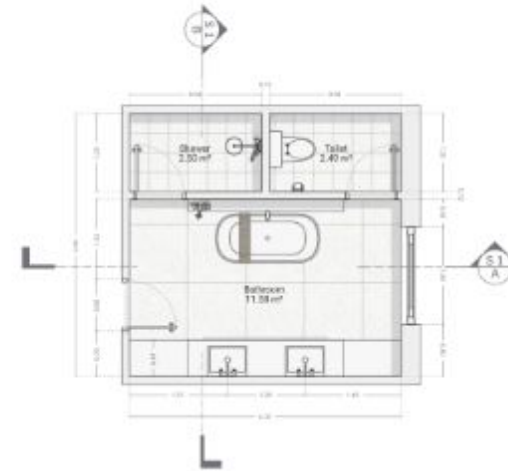
4,200 Gal.
Annual Water Saved / Home

Automated preference for KOHLER WaterSense® 1.28 GPF vs baseline 1.6 GPF fixtures.

100% SKU
Catalog Verification

Direct matching to verified KOHLER dimensional envelopes, finishes, and current pricing.

Executive Synthesis: KOHLER AI BathPlan transforms the highest-friction room in architecture into an explainable, sustainable, and confidence-driven design experience.



Deterministic Architectural Submittal

Code-checked layout with exact clearance envelopes and rough-in callouts ready for contractor build.

EVALUATION PACKAGE READY

Experience DesignPulse™ in the Live Working Prototype

Includes 2-minute walkthrough demo video, prompts documentation, and test suite.